

Lecture 4 p.1

Proof that there exists a unique soln. of

$$(*) \begin{cases} \dot{X} = V(X) \\ X(0) = X_0 \end{cases}$$

under assumption that $V(X)$ is Lipschitz with constant K , where $V: B_b(X_0) \rightarrow \mathbb{R}^n$.

Replace $(*)$ by its integral form:

$$(**) \quad X(t) = X_0 + \int_0^t V(X(s)) ds$$

[Soln. of $(**)$ satisfies $(*)$.]

Introduce following map T . — "Picard Iteration"

$$Tu = X_0 + \int_0^t V(u(s)) ds$$

If we have an initial $u_0(t)$, we can iterate this map to obtain a sequence of functions

$$u_0(t) \xrightarrow{T} u_1(t) = T(u_0(t)) \xrightarrow{T} u_2(t) = T(u_1(t)) \rightarrow$$

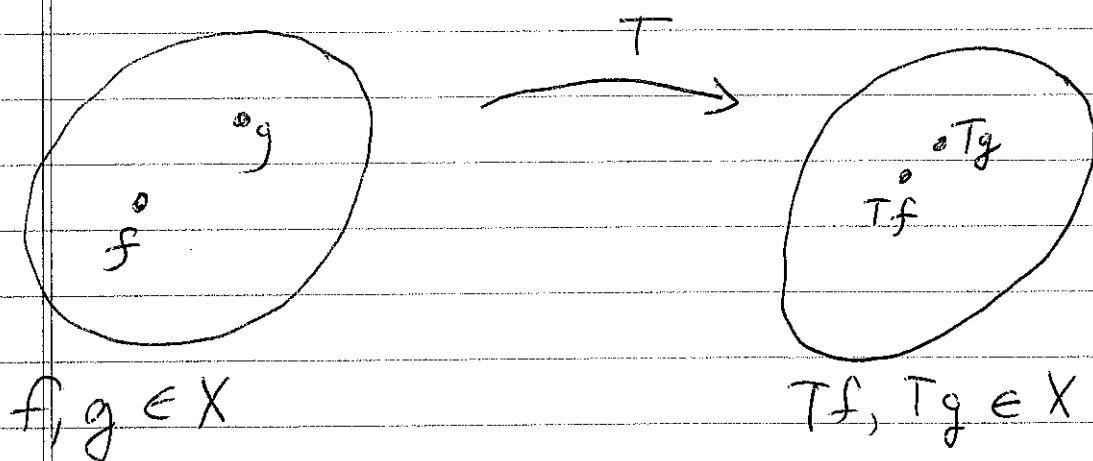
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If this iteration procedure converges to a fixed-pt. of the map, then we've found a soln. to $(**)$ and hence to $(*)$

Fixed-pt eqn: $TX(t) = X(t)$ is $(**)$:

$$X(t) = T(X(t)) = X_0 + \int_0^t V(X(s)) ds$$

The proof of existence-uniqueness thm. uses the contraction mapping thm — we will show that T is a contraction mapping, in which case it has a unique fixed pt.



X = "metric space"
(a set on which
we have a distance
function)

distance between Tf & Tg
smaller than distance
between f & g .

Contraction mapping Thm.

Let $T: X \rightarrow X$ be a map on a complete metric space X . The map is a contraction if there exists $c < 1$ s.t. for all $f, g \in X$

$$\rho(Tf, Tg) \leq c \rho(f, g)$$

In this case there is a unique fixed-pt.

We need a space X , a distance function $\rho(\cdot, \cdot)$, and then we need to show there is a $c < 1$.

$X = C^0(J, B_b(X_0))$ = set of continuous functions $u(t) \in B_b(X_0)$ for $t \in J = [-a, a]$

$$\text{let } \rho(u_1(t), u_2(t)) = \sup_{t \in J} |u_1(t) - u_2(t)|$$

sup = supremum = least upper bound

Note: $Tu(t)$ is continuous & stays in $B_b(X_0)$ via our restriction of $t \in J$.
Hence $T: X \rightarrow X$.

Finally, we need to find $c < 1$ s.t.

$$\rho(T(u_1(t)), T(u_2(t))) \leq c \rho(u_1(t), u_2(t))$$

$$\rho(T(u_1(t)), T(u_2(t))) = \sup_{t \in J} |Tu_1^{(t)} - Tu_2^{(t)}|$$

$\Rightarrow$

$$|Tu_1(t) - Tu_2(t)| = \left| \left(X_0 + \int_0^t V(u_1(s)) ds \right) - \left(X_0 + \int_0^t V(u_2(s)) ds \right) \right|$$

$$= \left| \int_0^t V(u_1(s)) - V(u_2(s)) ds \right|$$

$$\leq \int_0^t \underbrace{|V(u_1(s)) - V(u_2(s))|}_{\leq K |u_1(s) - u_2(s)|} ds$$

$$\leq K |u_1(s) - u_2(s)|$$

by Lipschitz condition

$$\leq K a \underbrace{\rho(u_1(t), u_2(t))}_{\sup_{t \in J} |u_1(t) - u_2(t)|}$$

$$\sup_{t \in J} |u_1(t) - u_2(t)|$$

$$\text{let } Ka = c < 1 \Rightarrow a < 1/K \text{ \& } a < b/M$$

(not in Thm. More work needed... See book)

Additional results in Chapter 3:

3.4 Dependence on initial conditions & parameters

3.5 Maximal Interval of Existence

Thm. 4.6 (Bounded Global Existence) If $V: \mathbb{R}^n \rightarrow \mathbb{R}$ is locally Lipschitz & bounded, then solu. of $\dot{X} = V(X)$, $X(0) = X_0$ exists for all $t \in \mathbb{R}$

Chapter 4:

A complete flow $\varphi_t(x)$ is a one-parameter differentiable mapping $\varphi: \mathbb{R} \times M \rightarrow M$ with following properties

$\begin{array}{ccc} & \uparrow & \nwarrow \uparrow \\ & \text{time} & \text{phase space} \end{array}$

(i) $\varphi_0(x) = x$

$\varphi_0 = \text{identity}$

(ii) $\varphi_t \circ \varphi_s = \varphi_{t+s}$

"group property"

all $t \in \mathbb{R}$, all $x \in M$

$$\begin{aligned} \varphi_t \circ \varphi_s(x) &= \varphi_t(\varphi_s(x)) = \varphi_{t+s}(x) \\ &= \varphi_s \circ \varphi_t(x) \end{aligned}$$

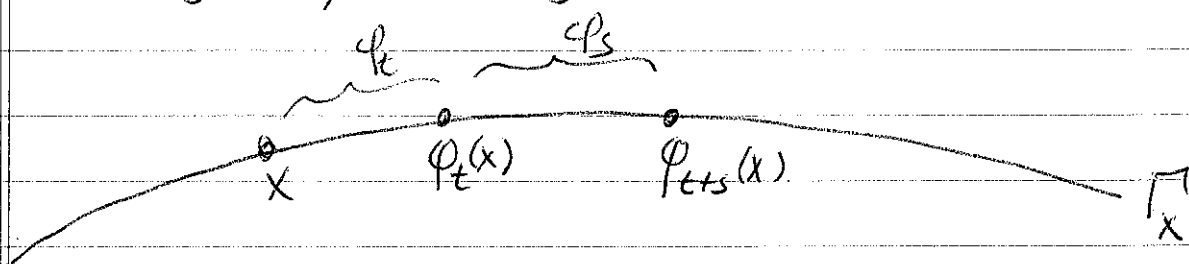
note φ_t is invertible

$$\varphi_{-t} \circ \varphi_t = \varphi_0$$

$$\varphi_t^{-1} = \varphi_{-t}$$

$\varphi_t(x)$ defines a curve Γ_x in M

as t varies over $\mathbb{R}$, called the orbit or trajectory through x

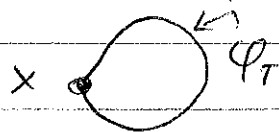


Examples

if $\varphi_t(x^*) = x^*$ for all t , then x^* is a fixed-pt of flow

if $\varphi_T(x) = x$ for some $T > 0$, where
 $\varphi_t(x) \neq x$ for $t \in (0, T)$

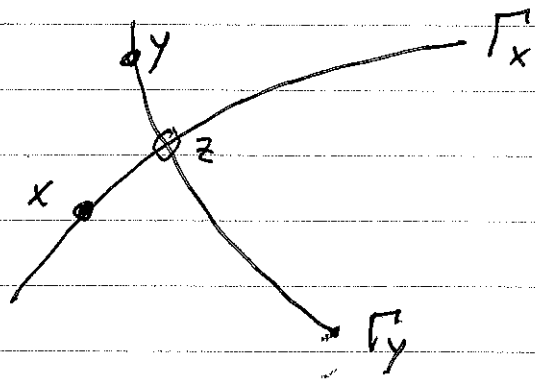
$\Rightarrow x$ is a periodic pt with period T



Show that $\varphi_s(x)$ is also a periodic pt with period T for any $s \in \mathbb{R}$

$$\varphi_T \circ \varphi_s(x) = \varphi_{T+s}(x) = \varphi_s \circ \varphi_T(x) = \varphi_s(x) \quad \checkmark$$

Group property $\varphi_t \circ \varphi_s = \varphi_{t+s}$ ~~implies~~ implies that two distinct trajectories cannot cross:



assume they cross at a pt. z

i.e. $\varphi_t(x) = z$ for some t

$\varphi_s(y) = z$ for some s

$$\Rightarrow \varphi_r(z) = \varphi_{t+r}(x) = \varphi_{s+r}(y) \quad \forall r \in \mathbb{R}$$

which contradicts assumption that they are distinct trajectories.

Connection with ODE's?

Flows are differentiable, so we can associate a vector field $V: M \rightarrow \mathbb{R}^n$, $n = \dim(M)$ with the flow

$$V(x) = \left. \frac{d}{dt} (\varphi_t(x)) \right|_{t=0}$$

$$= \lim_{\epsilon \rightarrow 0} \left(\frac{\varphi_{\epsilon}(x) - x}{\epsilon} \right)$$

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$\varphi_t(x)$ solves the following initial value problem

$$(*) \begin{cases} \frac{d}{dt} (\varphi_t(x_0)) = V(\varphi_t(x_0)) \\ \varphi_0(x_0) = x_0 \end{cases}$$

Check $\frac{d}{dt} (\varphi_t(x_0)) = \lim_{\epsilon \rightarrow 0} \left(\frac{\varphi_{t+\epsilon}(x_0) - \varphi_t(x_0)}{\epsilon} \right)$

$$= \lim_{\epsilon \rightarrow 0} \frac{\varphi_\epsilon(\varphi_t(x_0)) - \varphi_t(x_0)}{\epsilon}$$

$$= V(\varphi_t(x_0))$$